

Black's Zero-Beta Asset Pricing Model

Capital Market Equilibrium with Restricted Borrowing

1 Overview

The central contribution of Black's zero-beta asset-pricing model is that the standard linear relation between expected return and market beta can survive even when investors are unable to borrow freely at the risk-free rate.

The standard Sharpe–Lintner CAPM is

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f], \quad (1)$$

where

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}. \quad (2)$$

A key assumption behind this version of the CAPM is that investors are able to borrow and lend unlimited amounts at the same risk-free rate R_f .

Black shows that this assumption is **not necessary** for obtaining a linear beta–expected-return relation.

Black modifies the intercept of the CAPM, not the idea of linear beta pricing.

2 The Problem with Unrestricted Risk-Free Borrowing

In the standard CAPM, an investor who wants to hold a portfolio with a beta greater than one can borrow at the risk-free rate and invest more than 100 percent of wealth in the market portfolio.

For example,

$$R_p = (1 - w)R_f + wR_M, \quad w > 1. \quad (3)$$

The beta of this portfolio is

$$\beta_p = w. \quad (4)$$

Therefore, if $w = 1.5$, the investor effectively holds

150% in the market portfolio

and finances the additional investment by borrowing

50% at the risk-free rate.

In reality, however, individual investors generally cannot borrow unlimited amounts at the Treasury-bill rate. Black therefore studies capital-market equilibrium when borrowing at the risk-free rate is restricted or unavailable.

3 Black's Main Result

Black shows that expected returns remain linear in beta even without unrestricted risk-free borrowing.

The pricing equation becomes

$$\boxed{E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]} \quad (5)$$

where R_Z denotes the return on the **zero-beta portfolio**.

This equation is the central result of the Black zero-beta CAPM.

The main difference from the Sharpe–Lintner CAPM is therefore

$$\boxed{R_f \longrightarrow E(R_Z)} \quad (6)$$

in the intercept of the Security Market Line.

Thus,

The risk-free rate is replaced by the expected return of a zero-beta portfolio.

4 What Is the Zero-Beta Portfolio?

The zero-beta portfolio is a portfolio whose beta with respect to the market portfolio is equal to zero:

$$\beta_Z = \frac{\text{Cov}(R_Z, R_M)}{\text{Var}(R_M)} = 0. \quad (7)$$

Therefore,

$$\boxed{\text{Cov}(R_Z, R_M) = 0.} \quad (8)$$

An important distinction is that

$$\boxed{\text{zero beta} \neq \text{risk free.}} \quad (9)$$

The zero-beta portfolio may still have positive variance:

$$\text{Var}(R_Z) > 0. \quad (10)$$

Thus,

Zero beta means zero systematic market risk, not zero total risk.

The zero-beta portfolio may contain idiosyncratic risk, but its return has no systematic covariance with movements in the market portfolio.

5 Why Does $E(R_Z)$ Replace R_f ?

Consider the standard Sharpe–Lintner CAPM:

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f]. \quad (11)$$

Setting

$$\beta_i = 0$$

gives

$$E(R_i) = R_f. \quad (12)$$

Therefore, under the standard CAPM, the intercept of the Security Market Line is the risk-free rate.

When unrestricted risk-free borrowing and lending are unavailable, the relevant portfolio with zero market exposure is instead the zero-beta portfolio.

Therefore,

$$E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]. \quad (13)$$

At $\beta_i = 0$,

$$E(R_i) = E(R_Z). \quad (14)$$

Hence,

$E(R_Z)$ replaces R_f as the intercept of the Security Market Line.

6 Comparison with the Sharpe–Lintner CAPM

The standard Sharpe–Lintner CAPM is

$$\boxed{E(R_i) = R_f + \beta_i [E(R_M) - R_f]}. \quad (15)$$

The Black zero-beta CAPM is

$$\boxed{E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]}. \quad (16)$$

Both pricing relationships pass through the market portfolio:

$$\beta_M = 1, \quad (17)$$

and therefore

$$E(R_i) = E(R_M) \quad \text{when} \quad \beta_i = 1. \quad (18)$$

Under restricted borrowing, the expected return of the zero-beta portfolio can exceed the risk-free lending rate:

$$\boxed{E(R_Z) > R_f}. \quad (19)$$

When this occurs, the Black Security Market Line has

- a **higher intercept**, and
- a **smaller slope**.

The slope of the standard Sharpe–Lintner CAPM is

$$E(R_M) - R_f, \quad (20)$$

whereas the slope of the Black zero-beta CAPM is

$$E(R_M) - E(R_Z). \quad (21)$$

If

$$E(R_Z) > R_f,$$

then

$$E(R_M) - E(R_Z) < E(R_M) - R_f. \quad (22)$$

Therefore,

A higher zero-beta intercept implies a flatter Security Market Line.

Importantly,

A flatter SML does not mean that expected return decreases with beta.

The relation between beta and expected return can remain **positive**, but it is less steep than the Sharpe–Lintner CAPM predicts.

7 Economic Intuition for the Flatter Security Market Line

Suppose investors can borrow freely at R_f .

An aggressive investor who wants more systematic risk does not need to buy individual high-beta stocks. Instead, the investor can simply borrow money and buy more of the market portfolio.

For example,

$$150\% \text{ Market} - 50\% \text{ Risk-Free Asset} \tag{23}$$

produces approximately

$$\beta = 1.5. \tag{24}$$

Now suppose borrowing at the risk-free rate is prohibited or restricted.

Investors who want higher systematic risk can no longer simply lever the market portfolio. As a result, they have greater demand for high-beta securities.

Greater demand for high-beta securities increases their prices:

$$P_{\text{high-}\beta} \uparrow. \tag{25}$$

Higher prices imply lower expected returns:

$$E(R_{\text{high-}\beta}) \downarrow. \tag{26}$$

The resulting equilibrium therefore produces a flatter relation between beta and expected return.

Relative to the predictions of the Sharpe–Lintner CAPM,

Low-beta assets earn more than the Sharpe–Lintner CAPM predicts (27)

and

High-beta assets earn less than the Sharpe–Lintner CAPM predicts. (28)

This does **not** mean that low-beta assets necessarily earn higher returns than high-beta assets.

Instead,

Expected return may still rise with beta, but at a smaller rate.

This is one of the most important economic implications of Black’s model.

8 Two-Fund Separation

Another important result is that two-fund separation continues to hold even without unrestricted borrowing and lending at the risk-free rate.

Efficient portfolios can be represented as combinations of

1. the market portfolio M , and
2. the zero-beta portfolio Z .

For an efficient portfolio p ,

$$R_p = \beta_p R_M + (1 - \beta_p) R_Z. \quad (29)$$

Taking expectations gives

$$E(R_p) = \beta_p E(R_M) + (1 - \beta_p) E(R_Z) \quad (30)$$

$$= E(R_Z) + \beta_p [E(R_M) - E(R_Z)]. \quad (31)$$

Thus,

$$\boxed{E(R_p) = E(R_Z) + \beta_p [E(R_M) - E(R_Z)]}. \quad (32)$$

This shows how the linear beta-pricing relationship follows naturally from the two-fund representation.

9 Empirical Interpretation

The expected return–beta relation can generally be written as

$$E(R_i) = \gamma_0 + \gamma_1 \beta_i. \quad (33)$$

Under the Sharpe–Lintner CAPM,

$$\boxed{\gamma_0 = R_f} \quad (34)$$

and

$$\boxed{\gamma_1 = E(R_M) - R_f}. \quad (35)$$

The Black zero-beta CAPM instead implies

$$\boxed{\gamma_0 = E(R_Z)} \quad (36)$$

and

$$\boxed{\gamma_1 = E(R_M) - E(R_Z)}. \quad (37)$$

Therefore, Black's model does not require the intercept in a cross-sectional regression of expected returns on beta to equal the risk-free rate.

In particular, the model can accommodate

$$\gamma_0 > R_f \quad (38)$$

together with

$$\gamma_1 < E(R_M) - R_f. \quad (39)$$

This corresponds to an empirically **higher and flatter Security Market Line** than the strict Sharpe–Lintner CAPM predicts.

10 Connection to Early Empirical CAPM Evidence

Early empirical work on the CAPM found a pattern in which

- low-beta portfolios tended to earn **higher returns than predicted** by the Sharpe–Lintner CAPM, while
- high-beta portfolios tended to earn **lower returns than predicted** by the Sharpe–Lintner CAPM.

In other words, the empirical relation between expected returns and beta appeared approximately linear and positively sloped, but flatter than the standard CAPM predicted.

The Black zero-beta CAPM provides a theoretical explanation for this pattern.

The empirical relationship can take the form

$$E(R_i) = \gamma_0 + \gamma_1 \beta_i, \quad (40)$$

with

$$\gamma_0 > R_f \quad (41)$$

and

$$\gamma_1 < E(R_M) - R_f. \quad (42)$$

Therefore,

A Security Market Line that is too high and too flat

does not necessarily imply that beta pricing is completely wrong.

Instead, it may indicate that the Sharpe–Lintner assumption of unlimited borrowing at the risk-free rate is unrealistic.

11 Connection to BJS and Fama–MacBeth

The Black zero-beta model is especially important for interpreting the early empirical CAPM evidence.

BJS find that the empirical Security Market Line is approximately linear but **too high and too flat** relative to the strict Sharpe–Lintner CAPM.

This means

$$\gamma_0 > R_f$$

and

$$\gamma_1 < E(R_M) - R_f.$$

This is precisely the type of pattern that the Black zero-beta model can accommodate.

Fama and MacBeth similarly find evidence that the return–beta relation is approximately linear and positively sloped, but that the estimated intercept is above the risk-free rate:

$$\gamma_0 > R_f.$$

Thus,

The evidence rejects the strict Sharpe–Lintner intercept, but is more consistent with Black

The important distinction is therefore:

Linear beta pricing can survive even when the strict Sharpe–Lintner CAPM fails.

12 Main Difference Between the Two CAPM Versions

The central comparison can be summarized as follows:

	Sharpe–Lintner CAPM	Black Zero-Beta CAPM
Intercept	R_f	$E(R_Z)$
Slope	$E(R_M) - R_f$	$E(R_M) - E(R_Z)$
Return at $\beta = 0$	R_f	$E(R_Z)$
Risk-free borrowing	Required	Not required
Expected return–beta relation	Linear	Linear

13 The Key Equation to Remember

If only one equation is remembered from Black's model, it should be

$$\boxed{E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]} \quad (43)$$

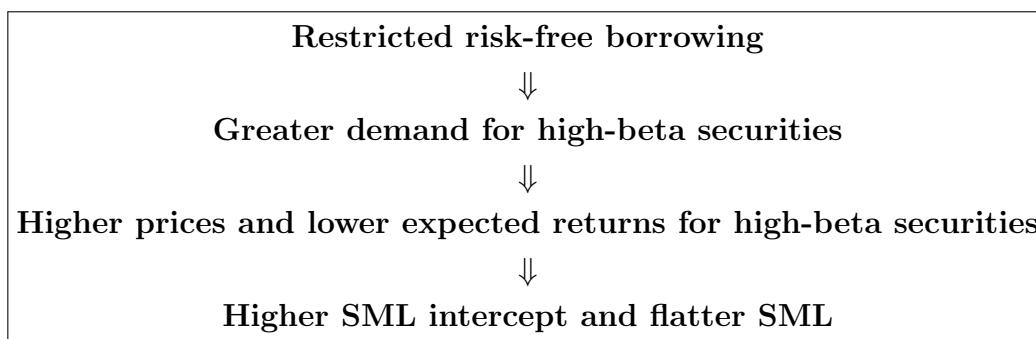
Compared with the Sharpe–Lintner CAPM,

$$\boxed{E(R_i) = R_f + \beta_i [E(R_M) - R_f]}, \quad (44)$$

the essential modification is

$$\boxed{R_f \longrightarrow E(R_Z)}. \quad (45)$$

The economic mechanism can be summarized as



14 Key Takeaways

- Black shows that unrestricted borrowing at the risk-free rate is **not necessary** for a linear relation between expected returns and market beta.
- When risk-free borrowing is restricted, the expected return of a **zero-beta portfolio** replaces the risk-free rate as the intercept of the Security Market Line.
- A zero-beta portfolio is **not necessarily risk-free**. It satisfies

$$\text{Cov}(R_Z, R_M) = 0,$$

but it may still have

$$\text{Var}(R_Z) > 0.$$

Thus, **zero beta means zero market risk, not zero total risk.**

- When

$$E(R_Z) > R_f,$$

the Black Security Market Line has a **higher intercept** and a **smaller slope** than the Sharpe–Lintner Security Market Line.

- Restricted borrowing increases investors’ demand for high-beta securities, raising their prices and reducing their expected returns relative to what the Sharpe–Lintner CAPM predicts.
- Therefore,

Low-beta stocks earn more than the strict CAPM predicts

while

High-beta stocks earn less than the strict CAPM predicts.

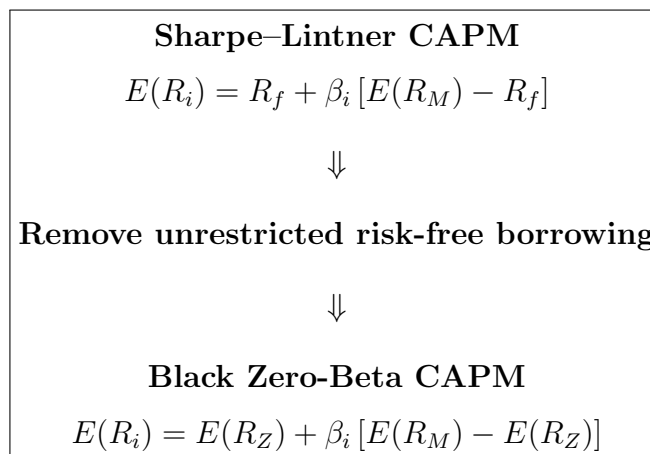
- This does **not** imply a negative return–beta relation. Expected return may still increase with beta, but the relation is **flatter** than the strict Sharpe–Lintner CAPM predicts.
- Black does **not reject beta pricing**. Instead, he modifies the intercept while preserving the linear beta-pricing relation.

15 One-Sentence Summary

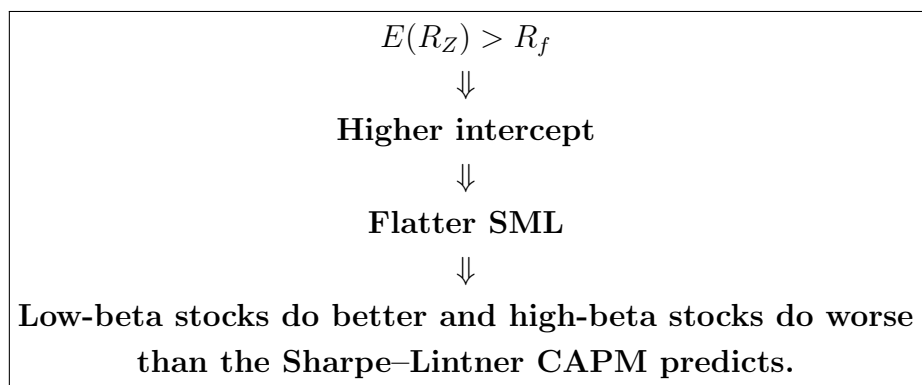
Black’s zero-beta CAPM shows that even when investors cannot borrow freely at the risk-free rate, equilibrium expected returns can remain linear in market beta, with the expected return on a zero-beta portfolio replacing the risk-free rate as the intercept of the Security Market Line.

16 Exam Summary

The entire model can be remembered through the following comparison:



Therefore,



The most important conceptual distinction is

Flatter SML $\neq$ negative return–beta relation.

Instead,

Return can still increase with beta, just less steeply than the Sharpe–Lintner CAPM predicts.
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